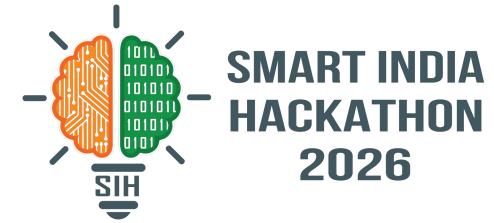


SMART INDIA HACKATHON 2026



- **Problem Statement ID** - 26038
- **Problem Statement Title**- Explainable AI for Diabetic Retinopathy Screening in Rural India
- **Theme**- MedTech / BioTech / HealthTech
- **PS Category**- Software
- **Team Name** - Optical-delusion



The problem

Ophthalmologist Shortage

Severe specialist scarcity relative to the diabetic population, especially in rural and semi-urban India.

Late Detection

Without on-site screening, DR advances to vision-threatening stages before patients seek specialist care.



The solution:

RetinaSense is a quality-aware, explainable DR screening and referral assistant for frontline health workers using low-cost fundus cameras. It produces a defensible, evidence-backed referral decision and not a black-box verdict.

What RetinaSense Does

- Accepts fundus images from low-cost cameras in field settings
- Validates image quality before analysis — prompts recapture if needed
- Detects microaneurysms and exudates with spatial localization
- Grades DR severity on the ICDR 0–4 scale with Grad-CAM attribution
- Outputs a referral recommendation with visible supporting evidence

What It Is Not

- Not a replacement for an ophthalmologist
- Not a black-box classifier with unexplained outputs
- Not designed for hospital-only deployment
- Not limited to high-quality clinical images only

RetinaSense supports the clinical decision that matters most: **who needs referral, and why.**

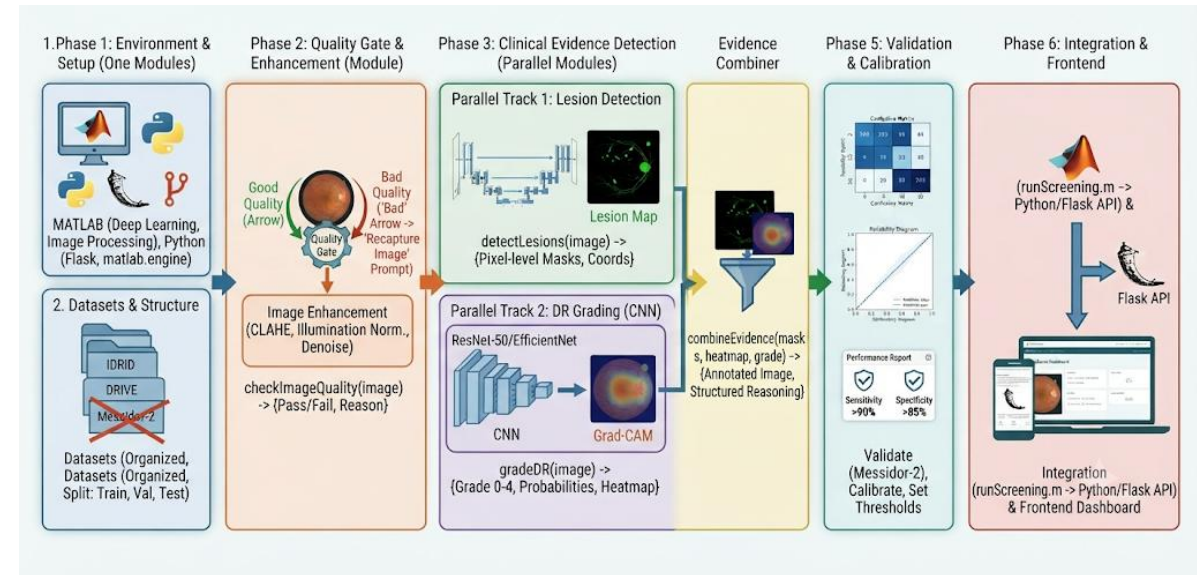
Why are we unique?

- **Recapture Feedback Loop:** Reflects real deployment conditions rare in academic DR projects. Unusable images trigger a recapture prompt rather than producing a forced, unreliable grade.
- **Evidence-Grounded Explainability:** Lesion detection and Grad-CAM attribution work in tandem, not attribution-only. Clinicians can verify both what was detected and where the model looked.
- **Referral-First Design:** Optimized for the clinical decision that matters, referral triage, not just fine-grained staging accuracy. Outputs a defensible "refer or don't" with supporting rationale.

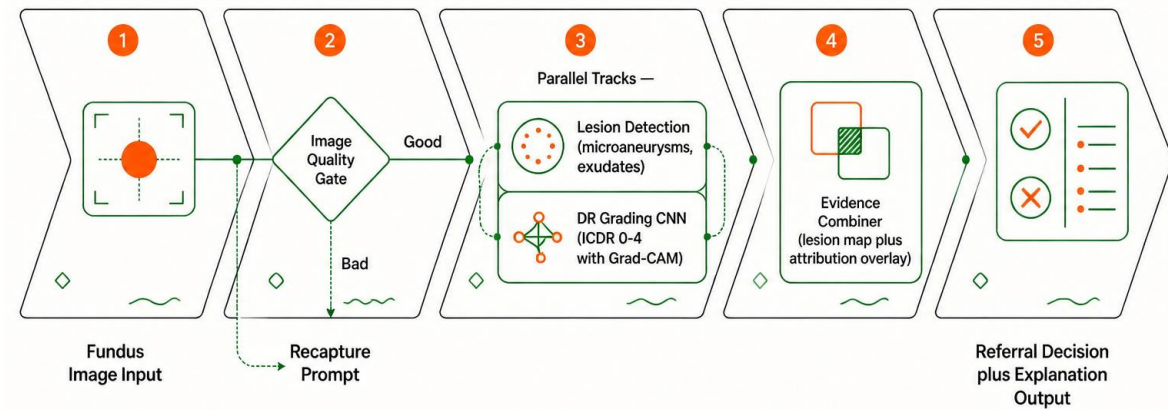
Technologies used:

- **MATLAB** – primary development platform
- **Image Processing Toolbox** – CLAHE, illumination normalization, denoising, morphological operations, vessel/optic-disc segmentation
- **Deep Learning Toolbox** – CNN architecture, transfer learning, pretrained network import
- **Computer Vision Toolbox** – image feature extraction, object/lesion detection support functions
- **gradCAM function** (native to Deep Learning Toolbox) – explainability/attribution overlay
- **Python + OpenCV/PyTorch**
- **Flask** – lightweight API server; receives image uploads, invokes MATLAB pipeline via matlab.engine, returns results as JSON
- **HTML / CSS / JavaScript** – base web stack
- **React** – component structure: Upload → Quality Check → Result Dashboard
- **ONNX** – interchange format, used to import any externally pretrained model into MATLAB's Deep Learning Toolbox
- **Git/GitHub** – version control and team collaboration
- **DATASETS**
 - ✓ APTOS 2019 (Kaggle) – grading
 - ✓ IDRiD (IEEE DataPort) – lesion segmentation + grading
 - ✓ DRIVE (Grand Challenge) – vessel segmentation
 - ✓ Messidor-2 (ADCIS) – external validation

Implementation:



Working of the final product:



Why It's Feasible

- Public benchmark datasets provide grading, lesion, and vessel annotations
- Established techniques throughout — transfer learning, CLAHE, Grad-CAM — no unproven research
- MATLAB's pretrained-model import avoids training a CNN from scratch

Phase 2 Roadmap (Post-MVP)

- Hemorrhage and neovascularization detection
- Full 5-level ICDR grading with calibrated confidence

Risks & How We Address Them

Cross-Dataset Generalization

APTOS for development; Messidor-2 as explicit external validation — reported honestly, not hidden.

Class Imbalance

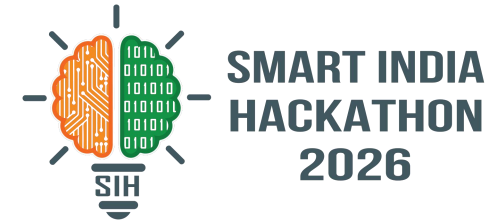
Class-weighted loss and augmentation for underrepresented severe/proliferative cases.

Scope Management

MVP scoped to 2 lesion types (microaneurysms + exudates); remaining lesion types deferred to Phase 2.

Optical-delusion

IMPACT AND BENEFITS



Potential impact on the target audience

Supports earlier identification of patients who may require ophthalmic evaluation, potentially enabling treatment before disease progresses to vision-threatening stages. Extends frontline workers' screening and referral support capability without requiring specialist presence.

BENEFITS INCLUDE:



Social Equity

Improves healthcare equity between rural and urban diabetic populations by enabling on-site screening without specialist presence.



Economic Efficiency

Early screening is significantly cheaper than treating advanced DR or managing permanent vision loss — reducing long-term system costs.



Health System Relief

Pre-filters which patients genuinely need specialist referral, easing pressure on scarce ophthalmologist capacity in underserved regions.



Clinician Trust

Evidence-grounded explainability lets clinicians verify rather than blindly trust the output — critical for adoption in real clinical workflows.



Clinical Standard

International Clinical Diabetic Retinopathy (ICDR) Severity Scale

— the globally recognized 5-level grading framework (Level 0: No DR through Level 4: Proliferative DR). RetinaSense outputs are calibrated to this standard.

Key References

- **Explainability:** Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks," ICCV 2017
- **Implementation:** MATLAB Deep Learning Toolbox — `gradCAM` function, pretrained network import
- **Image Enhancement:** CLAHE (Contrast-Limited Adaptive Histogram Equalization) for low-quality fundus image normalization

Datasets Used

APTOS 2019	Grading training (no lesion masks)
IDRiD	Lesion segmentation + grading (Indian-specific)
DRIVE	Vessel pretraining only
Messidor-2	External validation (grading only)